Tactix: A One-Reason Reproducer for H2

Report to Andrew

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Outcome

The outstanding native-DSR H2 checker rejection now has a substantially smaller, pinned reproducer: 16 original-CNF clauses and a 314-byte proof, down from 42 clauses and 680 bytes. The same public-API action program still returns UNSAT. A clause-deletion shrink preserved the exact checker-contract split across 88 candidate tests in hwulab SLURM job `1723448`, which completed successfully in three seconds. The final case is one-clause- deletion minimal under this specific oracle, not a general minimal proof. Full hashes, commands, logs, and artifacts are in AM436.

Certificate evidence

The frozen canonical no-option `dsr-trim` rejects the reduced proof in backward mode at addition line 21 (a watch-pointer error) and in forward mode at line 12 (no UP contradiction for RAT clause 15). Its verbose backward trace identifies just one ignored deletion: input clause 15, `-18 19`, is retained as an implied-unit reason. With the diagnostic `-delete-implied-units` option, both backward and forward elaborations return `s VERIFIED UNSAT`; pinned Trestle independently verifies each decoded LSR against the same original CNF. This isolates the proof-state boundary but does *not* justify changing the production checker option.

A negative control retaining only the first proof line returns exit code zero with `s VALID`, not `s VERIFIED UNSAT`. It confirms that the checker status, not process exit code alone, must be the acceptance test. The current production oracle already enforces that distinction.

Next actions

Use the one-reason case to test a general checker-visible proof-state model or an independent derivation that survives the canonical no-option checker. Do not mask the public SR action or patch only this proof. The separate AES extended-checker array `1723443` and full Debug job `1723391` remain running on their frozen pins. H2/N2, broad N1, N3/N4 and retained performance gates are open. The production online-learning refactors and the requested full 5,000-second Tier-D comparison remain downstream of those correctness and performance gates.